AquaVision-X: A Multimodal CNN–XAI Framework for Marine Plastic Waste Detection Using Drone-Based Aerial Imagery

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Submitted by

P. Jagruthi	**23881A05BA**
B. Sai	**23881A05Y7**
Md. Sohail	**23881A05AT**

SUPERVISOR

Mr. Tummala Vinay Goud

Assistant Professor

VARDHAMAN COLLEGE OF ENGINEERING

Department of Computer Science and Engineering

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VARDHAMAN
COLLEGE OF ENGINEERING

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the mini-project titled **AquaVision-X: A Multimodal CNN–XAI Framework for Marine Plastic Waste Detection Using Drone-Based Aerial Imagery** is carried out by

P. Jagruthi	**23881A05BA**
B. Sai	**23881A05Y7**
Md. Sohail	**23881A05AT**

in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Computer Science and Engineering** during the year 2025-26.

Signature of the Supervisor
Mr. Tummala Vinay Goud
Assistant Professor of CSE

Signature of the HOD
Dr. S.V.Vasantha
Associate Professor and HOD, CSE

Signature of the External Examiner

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P. Jagruthi

B. Sai

Md. Sohail

Abstract

Marine Plastic Pollution represents a significant global environmental problem and requires effective, scalable ways to monitor it. To address this problem, we present AquaVision-X, a multimodal Deep Learning Framework designed for the identification of marine plastic debris located by aerial drone imagery. AquaVision-X consists of an end-to-end Convolutional Neural Network (CNN) architecture combined with Explainable Artificial Intelligence (XAI) in order to provide both high levels of accuracy and interpretability when detecting marine plastics. By using multimodal inputs, which combine visual images with contextual metadata, we have improved the reliability of our Model across multiple environmental factors, including water conditions, lighting environment, and shoreline location. Our user interface allows for easy uploading of images, rapid, automated detection of marine plastic debris, and visualization of bounding boxes associated with XAI-based heatmaps indicating the features that influenced the models prediction. Testing results demonstrated that AquaVision-X has excellent detection performance, as well as excellent transferability to real-world images of coastal locations. AquaVision-X provides a scalable, interpretable, and actionable solution for marine conservation efforts, and in the future it could be applied to detection of underwater plastic using Multispectral Sensing.

Keywords: Marine Plastic Pollution, Deep Learning, Convolutional Neural Networks (CNN), Multimodal Learning, Aerial Drone Imagery, Explainable Artificial Intelligence (XAI), Environmental Monitoring, Plastic Debris Detection, Computer Vision, Coastal Conservation, Bounding Box Detection, Heatmap Visualization, Multispectral Sensing..

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Abbreviations

Abbreviation	Description
XAI	Explainable Artificial Intelligence
CNN	Convolutional Neural Network
MLP	Multilayer Perceptron
DL	Deep Learning
AI	Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
ReLU	Rectified Linear Unit
API	Application Programming Interface
GUI	Graphical User Interface
ML	Machine Learning
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
UI	User Interface
UX	User Experience
F1-Score	Harmonic Mean of Precision and Recall
Precision	Ratio of Correct Positive Predictions
Recall	Ratio of Actual Positives Correctly Identified

CHAPTER 1

Introduction

1.1 Introduction

Marine plastic pollution has emerged as one of the most critical environmental challenges worldwide, posing severe threats to marine ecosystems, biodiversity, and human health. Millions of tons of plastic waste enter oceans every year, affecting marine organisms through ingestion, entanglement, and habitat disruption. The persistence of plastic materials in water bodies leads to long-term ecological imbalance and contributes to the growing global environmental crisis.

Traditional methods for monitoring marine plastic waste primarily rely on manual surveys and field inspections, which are labor-intensive, time-consuming, and limited in coverage. These approaches are not suitable for large-scale monitoring of oceans and coastal regions. Therefore, there is a pressing need for automated, scalable, and efficient solutions that can accurately detect and track marine debris in real time.

Recent advancements in Artificial Intelligence (AI) and Deep Learning (DL) have opened new possibilities for addressing environmental challenges. In particular, Convolutional Neural Networks (CNNs) have shown exceptional capabilities in image classification and object detection tasks. By leveraging these techniques.

In this project, we propose **AquaVision-X**, a multimodal deep learning framework for marine plastic waste detection using drone-based aerial imagery. The proposed system not only relies on visual image data but also incorporates contextual and environmental information such as lighting conditions, water clarity, and geographical features. Explainable Artificial Intelligence (XAI) techniques are integrated into the framework. Specifically, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to generate visual explanations in the form of heatmaps, which highlight the important regions of the image influencing the model's predictions. This improves transparency and builds trust in the system. Furthermore, AquaVision-X is designed to be scalable and adaptable, enabling its application across different coastal environments and varying environmental conditions. The integration of advanced deep learning techniques with explainability ensures both high performance and interpretability.

In conclusion, this project presents an efficient, reliable, and interpretable solution for marine plastic waste detection. By combining CNN-based detection, multimodal learning, and XAI techniques, AquaVision-X contributes towards sustainable environmental monitoring and supports global efforts in marine conservation.

1.2 Background and Motivation

Marine environments across the globe are increasingly threatened by the accumulation of plastic waste. It is estimated that millions of tons of plastic enter the oceans annually, leading to severe consequences for marine life, ecosystems, and coastal economies. Plastics are non-biodegradable and can persist in the environment for hundreds of years, breaking down into microplastics that further contaminate water bodies and enter the food chain. This widespread pollution not only affects aquatic organisms but also poses significant risks to human health and environmental sustainability.

Conventional methods for detecting and monitoring marine plastic debris include manual surveys, satellite imaging, and field observations. While these approaches provide valuable insights, they suffer from several limitations such as high cost, limited spatial and temporal coverage, and dependency on human effort. Moreover, manual analysis of large volumes of image data is prone to errors and inefficiencies, making it unsuitable for real-time monitoring applications.

In recent years, advancements in Machine Learning (ML) and Deep Learning (DL) have revolutionized the field of computer vision, enabling automated analysis of complex visual data. Convolutional Neural Networks (CNNs), in particular, have demonstrated superior performance in image classification and object detection tasks. These models can learn hierarchical features from images, making them highly effective in identifying patterns such as marine plastic debris in aerial imagery.

Despite these advancements, a major challenge in deploying deep learning models is the lack of transparency and interpretability. Most models operate as "black boxes," making it difficult to understand how decisions are made. This lack of explainability can reduce user trust, especially in critical applications such as environmental monitoring and policy-making. To address this issue, Explainable Artificial Intelligence (XAI) techniques have been introduced to provide insights into model behavior.

Motivated by these challenges, this project aims to develop an intelligent and interpretable system for marine plastic waste detection. The proposed AquaVision-X framework integrates CNN-based image analysis with multimodal data and XAI techniques to enhance both accuracy and transparency. By combining visual and contextual information, the system is designed to perform reliably under varying environmental conditions.

Overall, this project is driven by the need for an automated, efficient, and interpretable approach to tackle the growing problem of marine plastic pollution using modern Artificial Intelligence technologies.

1.3 Objectives of the Project Work

The main objective of this project is to develop an advanced, accurate, and interpretable system for detecting marine plastic waste using drone-based aerial imagery and deep learning techniques. The project aims to address the limitations of traditional monitoring methods by providing an automated and scalable solution. The detailed objectives of the project are as follows:

- To design and develop a robust Convolutional Neural Network (CNN) model capable of identifying and localizing marine plastic waste in aerial images with high accuracy.

- To utilize drone (UAV)-based imaging for collecting high-resolution data, enabling efficient monitoring of large coastal and marine areas.

- To develop a multimodal learning framework that integrates visual image data with contextual and environmental factors such as lighting conditions, water clarity, and geographical variations to improve detection performance.
- To enhance the generalization capability of the model so that it performs reliably across diverse environmental conditions and different types of marine debris.
- To incorporate Explainable Artificial Intelligence (XAI) techniques, specifically Gradient-weighted Class Activation Mapping (Grad-CAM), to provide visual explanations of model predictions and improve transparency.
- To develop a scalable and efficient system that can be extended for continuous monitoring and integrated with real-time drone systems.
- To contribute towards environmental sustainability by providing a technological solution that aids in marine pollution monitoring and management.

1.4 Organization of the Report

This report is organized in a structured manner to provide a clear understanding of the proposed AquaVision-X system and its components. Each chapter presents a specific aspect of the project, covering the problem, methodology, implementation, and results.

- **Chapter 1: Introduction**
  This chapter introduces the problem of marine plastic pollution and its environmental impact. It outlines the motivation, objectives, and need for an automated detection system.
- **Chapter 2: Literature Survey**
  This chapter reviews existing research related to marine debris detection and deep learning techniques. It highlights the limitations of current approaches and identifies the research gap.

- **Chapter 3: Proposed Methodology**
  This chapter explains the working of the AquaVision-X system, including CNN-based detection, multimodal learning, and the use of Grad-CAM for explainability.

- **Chapter 4: System Implementation**
  This chapter describes data collection, preprocessing, model training, and the development of the user interface for detection and visualization.

- **Chapter 5: Results and Discussion**
  This chapter presents experimental results, evaluation metrics, and analysis of system performance, along with observations and limitations.

- **Chapter 6: Conclusion and Future Scope**
  This chapter summarizes the project outcomes and suggests possible improvements such as real-time monitoring and advanced detection techniques.

- **References and Appendices**
  This section includes all referenced materials and additional supporting content such as outputs and system details.

CHAPTER 2

Literature Survey

2.1 Introduction

The rapid growth of marine plastic pollution has attracted significant attention from researchers across the globe, leading to the development of various techniques for detecting and monitoring marine debris. In recent years, advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have enabled automated analysis of large-scale environmental data, particularly in the field of image processing and computer vision.

This chapter presents a comprehensive review of existing research works related to marine plastic waste detection using aerial imagery, satellite data, and deep learning techniques. Several approaches have been proposed to address this problem, ranging from traditional image processing methods to advanced Convolutional Neural Network (CNN)-based models.

Early methods for marine debris detection primarily relied on manual inspection and classical image processing techniques such as thresholding, edge detection, and feature extraction. Although these methods provided initial insights, they were limited in terms of accuracy, scalability, and adaptability to varying environmental conditions such as lighting changes, water reflections, and background noise.

Explainable Artificial Intelligence (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) have been introduced. These techniques provide visual explanations, helping users interpret model decisions and increasing trust in the system. In summary, the literature indicates a clear shift from traditional methods to advanced deep learning-based approaches for marine plastic detection. The proposed AquaVision-X system aims to address these gaps by integrating CNN-based detection with multimodal learning and XAI techniques to provide an accurate, scalable, and interpretable solution.

2.2 Review of Prior Research

In recent years, several research works have been carried out to address the problem of marine plastic waste detection using advanced image processing and deep learning techniques. This section presents a review of some significant contributions in this domain.

Azhan Mohammed (2022) proposed a deep learning model called ResAttUNet for detecting marine debris. This model integrates residual connections with attention mechanisms to improve feature extraction and segmentation performance. The attention module helps the network focus on relevant regions in the image, thereby enhancing detection accuracy. However, the model primarily focuses on segmentation and lacks interpretability features to explain its predictions.

M. Srinivasa Rao (2025) introduced a hybrid deep learning approach for marine debris detection in satellite imagery using a UNet architecture with a ResNeXt50 backbone. This method improves feature representation and achieves better accuracy in detecting debris across large areas. While the approach is effective for satellite images, it may not perform well for high-resolution drone imagery and does not incorporate explainable AI techniques.

Bhanumathi M. et al. (2022) proposed a deep learning-based approach for marine plastic detection, emphasizing the use of convolutional neural networks for classification tasks. Their work highlights the potential of CNN models in identifying plastic waste in images. However, the approach does not incorporate multimodal data or provide insights into model decision-making.

The proposed AquaVision-X system addresses these gaps by integrating CNN-based detection with multimodal inputs and Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM. This combination not only improves detection accuracy but also provides transparency in model predictions, making the system more reliable and practical for real-world applications.

Table 2.1: Comparison of Existing and Proposed Techniques

Ref	Technology	Data	Contribution
[1]	CNN (Deep Learning)	Marine Images	Basic plastic classification
[2]	CNN + Bounding Boxes	Drone Images	Object detection and localization
[3]	Multimodal Learning	Image + Context	Improved detection using additional data
[4]	CNN (Deep Learning)	Coastal Images	Large-scale debris detection
[5]	XAI (Grad-CAM)	Marine Dataset	Visual explanation of predictions
Proposed	CNN + Multimodal + XAI	Drone + Context	Accurate, explainable, and localized detection

2.3 Identified Research Gaps

Based on the analysis of existing studies in marine plastic detection, several key research gaps have been identified. While deep learning techniques such as CNNs have improved detection accuracy, many systems still lack important features required for practical applications.

- **Lack of Explainability**
  Most CNN-based models act as black-box systems, providing predictions without explaining how decisions are made. This reduces transparency and limits user trust in the system.

- **Limited Use of Multimodal Information**
  Many existing approaches rely only on image data for detection. However, real-world conditions such as lighting, water clarity, and environmental context are not considered, which affects model performance.

- **Inadequate Object Localization**
  Some models focus only on classification and do not clearly identify the exact location of plastic waste. The absence of bounding box-based localization reduces the practical usefulness of these systems.

- **Poor Generalization**
  Models trained on specific datasets often fail to perform well under different environmental conditions, limiting their real-world applicability.

- **Limited Integration of XAI Techniques**

  Although Explainable AI methods like Grad-CAM exist, they are not widely integrated into marine detection systems. This creates a gap in interpretability and validation of model predictions.

To overcome these limitations, the proposed AquaVision-X system integrates CNN-based deep learning with multimodal data and XAI techniques such as Grad-CAM. It also incorporates bounding box-based detection for accurate localization, resulting in a more reliable, interpretable, and practical solution.

2.4 Summary

This chapter presented a detailed review of existing research works and methodologies related to the detection of marine plastic waste using advanced computational techniques. Various approaches, including traditional image processing methods, machine learning algorithms, and deep learning models such as Convolutional Neural Networks (CNNs) and UNet-based architectures, have been critically analyzed.

The literature survey clearly indicates a significant shift from conventional manual and rule-based methods to automated deep learning-based solutions. CNN-based models have demonstrated strong performance in image classification and object detection tasks, enabling accurate identification of marine debris. Similarly, UNet and its variants have been effectively utilized for segmentation tasks, allowing precise localization of plastic waste in images.

Despite these advancements, several critical challenges remain. Most existing systems rely on single-modal data, limiting their ability to adapt to varying environmental conditions such as lighting variations, water reflections, and complex backgrounds. Additionally, the majority of deep learning models operate as black-box systems, lacking transparency and interpretability. This makes it difficult for users to understand the reasoning behind predictions, thereby reducing trust in the system.

Another key limitation observed is the lack of real-time implementation and scalability in many research works. While models achieve good performance in controlled environments, their deployment in real-world scenarios remains a challenge due to computational constraints and variability in input data. Furthermore, many systems do not provide user-friendly interfaces, restricting their usability for non-technical stakeholders such as environmental agencies.

To address these limitations, the proposed AquaVision-X framework introduces an integrated approach that combines CNN-based detection with multimodal data and Explainable Artificial Intelligence (XAI) techniques. By incorporating contextual information along with visual data, the system improves detection accuracy and robustness. The use of Grad-CAM enables the generation of visual heatmaps, providing insights into the model's decision-making process and enhancing interpretability.

Moreover, the proposed system focuses on developing a user-friendly interface that supports real-time detection and visualization, making it practical for deployment in real-world applications. The integration of advanced deep learning techniques with explainability ensures that the system is not only accurate but also reliable and transparent.

In conclusion, this chapter highlights the strengths and limitations of existing approaches and establishes the need for a comprehensive solution that addresses accuracy, interpretability, scalability, and usability. The insights gained from the literature survey form the foundation for the design and development of the proposed AquaVision-X system, which aims to contribute towards effective marine pollution monitoring and environmental conservation.

CHAPTER 3

Analysis of Existing Marine Plastic Detection Techniques

3.1 Introduction

This chapter presents a detailed analysis of existing techniques used for marine plastic waste detection and related environmental monitoring applications. Marine plastic pollution has become a critical global issue, necessitating the development of efficient, scalable, and automated detection systems. In recent years, the rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) has led to the emergence of various techniques for identifying marine debris from image data. These approaches have significantly improved detection accuracy and reduced the dependency on manual inspection methods.

The primary focus of this chapter is on deep learning-based models, particularly Convolutional Neural Networks (CNNs), which have been widely adopted for image classification, object detection, and segmentation tasks. CNN-based models have demonstrated the ability to automatically extract hierarchical features from images, making them highly effective in detecting complex patterns such as marine plastic waste. Various architectures, including standard CNNs, UNet, and hybrid models, are analyzed to understand their performance, advantages, and limitations.

Another key aspect discussed in this chapter is the integration of multimodal data. While most existing systems rely solely on visual image data, environmental factors such as lighting conditions, water surface variations, weather conditions, and geographical features play a crucial role in detection accuracy. The lack of integration of such contextual information in many existing models limits their robustness in real-world scenarios. Therefore, multimodal approaches are analyzed as a potential solution to improve system performance.

Furthermore, this chapter explores the importance of Explainable Artificial Intelligence (XAI) in marine plastic detection. Although deep learning models achieve high accuracy, they often function as black-box systems, making it difficult to interpret their decisions. This lack of transparency poses challenges in practical deployment, especially for environmental monitoring applications where reliability and trust are essential. XAI techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) are studied to understand how they provide visual explanations by highlighting important regions in the input image, thereby improving interpretability.

Furthermore, this chapter explores the importance of Explainable Artificial Intelligence (XAI) in marine plastic detection. Although deep learning models achieve high accuracy, they often function as black-box systems, making it difficult to interpret their decisions. This lack of transparency poses challenges in practical deployment, especially for environmental monitoring applications where reliability and trust are essential.

XAI techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) are studied to understand how they provide visual explanations by highlighting important regions in the input image, thereby improving interpretability. The chapter also discusses key performance factors such as accuracy, generalization capability, computational efficiency, and scalability. Many existing systems perform well.

3.2 Image-Based Marine Plastic Detection Systems

Image-based marine plastic detection systems are among the most widely used approaches in environmental monitoring applications. These systems utilize aerial or satellite images to identify plastic waste using machine learning and deep learning techniques. Convolutional Neural Networks (CNNs) have proven to be highly effective in extracting features such as texture, shape, color variations, and spatial patterns from images. These features enable accurate detection and classification of marine plastic debris in different environmental conditions.

Several researchers have utilized publicly available datasets and collected image data from satellite and drone-based platforms to train their models. Satellite imagery provides large-scale coverage of ocean and coastal regions, while drone-based imagery offers high-resolution data for localized detection. Although these datasets are useful for training deep learning models, they often fail to capture the full variability of real-world conditions such as lighting variations, water reflections, waves, and background noise. As a result, models trained on such datasets may not generalize well in practical scenarios.

To improve detection performance, techniques such as data augmentation and transfer learning have been widely adopted. Pre-trained models like ResNet, EfficientNet, and other CNN architectures are commonly used to reduce training time and enhance feature extraction capabilities. These approaches help in improving accuracy and handling limited datasets. However, most of these methods rely solely on visual image data, which restricts their ability to consider environmental factors that influence detection performance.

Furthermore, image-based systems often face challenges in distinguishing plastic waste from visually similar objects such as sea foam, shadows, or natural debris. This can lead to false detections and reduced reliability of the system. In addition, many existing models operate as black-box systems without providing explanations for their predictions, which limits their usability in real-world environmental monitoring. These limitations are addressed in the proposed AquaVision-X framework through the use of multimodal learning and Explainable Artificial Intelligence (XAI) techniques.

3.3 Explainable AI in Marine Plastic Detection

Explainable Artificial Intelligence (XAI) has gained significant importance in recent years due to the increasing adoption of deep learning models in environmental monitoring applications. While deep learning techniques such as Convolutional Neural Networks (CNNs) provide high accuracy in detecting marine plastic waste, they often operate as black-box systems. This makes it difficult to understand how predictions are generated, thereby reducing user trust and limiting their practical applicability in real-world scenarios.

In the context of marine plastic detection, interpretability is crucial, as environmental agencies and researchers need to verify whether the system is correctly identifying plastic waste. Without proper explanations, it becomes challenging to rely on automated systems for decision-making processes such as pollution monitoring and cleanup planning. Therefore, integrating explainability into detection systems is essential to improve transparency and reliability.

One of the most widely used techniques in Explainable AI is Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM generates visual heatmaps that highlight the important regions in an image contributing to the model's prediction. In marine plastic detection, these heatmaps indicate the specific areas in drone or satellite images where plastic waste is detected. This allows users to visually validate whether the model is focusing on relevant features such as plastic debris instead of irrelevant background elements like water reflections or shadows. In conclusion, Explainable AI plays a vital role in improving the usability, transparency, and trustworthiness of marine plastic detection systems. By providing visual explanations of model predictions, XAI bridges the gap between complex deep learning models and end users. The proposed AquaVision-X system incorporates Grad-CAM-based explanations to ensure that the detection results are not only accurate but also interpretable, making the system more reliable and suitable for practical deployment.

3.4 Multimodal Learning Approaches in Marine Plastic Detection

Multimodal learning is an emerging approach that combines multiple types of data to improve the performance and robustness of deep learning models. In the context of marine plastic detection, multimodal learning involves integrating visual image data with contextual and environmental information such as lighting conditions, water clarity, weather conditions, and geographical location. These factors significantly influence the visibility and detection of plastic waste in marine environments. By incorporating such information, models can make more accurate and context-aware predictions.

Several recent studies have explored the integration of environmental data with image-based deep learning models for marine monitoring applications. These approaches typically use Convolutional Neural Networks (CNNs) to extract features from images and additional neural networks, such as Multilayer Perceptrons (MLPs), to process non-visual data. The extracted features from different modalities are then combined using feature fusion techniques. This integration helps in improving detection accuracy and enhances the model's ability to handle complex environmental variations. Another significant challenge is the variability of marine environments. Factors such as water reflections, waves, lighting conditions, and background noise can affect both image quality and environmental data consistency. This makes it difficult to design models that generalize well across different conditions. Additionally, the availability of datasets that include both high-quality images and corresponding environmental metadata is limited, which further restricts the development of robust multimodal systems.

The proposed AquaVision-X system leverages multimodal learning by integrating drone-based image data with contextual environmental information. This improves detection accuracy, robustness, and generalization capability. By combining multimodal learning with Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM, the system not only enhances performance but also provides interpretable results, making it suitable for practical deployment in marine environmental monitoring applications.

3.5 Need for Multimodal XAI-Based AquaVision-X Framework

Marine environments are influenced by multiple dynamic factors such as water conditions, lighting variations, weather patterns, and geographical characteristics. Traditional methods for detecting marine plastic waste rely on manual observation and field surveys, which are time-consuming, labor-intensive, and limited in coverage. With the advancement of deep learning, image-based detection systems have been developed to automate this process. However, most existing systems depend solely on visual data, which limits their effectiveness in complex and varying real-world marine environments.

The detection of marine plastic waste is highly influenced by environmental factors such as water clarity, sunlight reflection, wave patterns, and background noise. For instance, strong sunlight reflections or waves can obscure plastic objects, making them difficult to detect using image-only models. Similarly, variations in water color and coastal features can lead to false detections.

The integration of Explainable Artificial Intelligence (XAI) techniques, such as Gradient-weighted Class Activation Mapping (Grad-CAM), addresses this issue by generating visual heatmaps that highlight the regions of the image influencing the model's prediction. This enables users to verify whether the system is correctly identifying marine plastic waste. The proposed AquaVision-X framework addresses this limitation by providing precise localization through bounding boxes and visual explanations, enabling better monitoring and management of marine pollution.

Furthermore, the increasing demand for sustainable environmental solutions highlights the need for intelligent, scalable, and user-friendly systems. A multimodal XAI-based framework not only improves detection accuracy but also enhances usability through interpretability and real-time visualization. Such a system can be integrated with drone technology and deployed in coastal regions for continuous monitoring of marine environments.

Overall, the need for the AquaVision-X framework arises from the limitations of existing image-based approaches and the growing demand for reliable, transparent, and scalable marine plastic detection systems. By combining multimodal learning with Explainable Artificial Intelligence, the proposed system aims to provide an accurate, interpretable, and practical solution for marine environmental monitoring and conservation efforts.

3.6 Summary

This chapter provided a comprehensive analysis of existing techniques used in marine plastic waste detection and related environmental monitoring applications. It covered key approaches such as image-based detection systems, Explainable Artificial Intelligence (XAI) techniques, and multimodal learning methods. Each approach was critically analyzed in terms of its methodology, advantages, and limitations, providing a clear understanding of the current state of research in this domain.

Image-based detection systems were found to be effective in identifying marine plastic waste using deep learning models such as Convolutional Neural Networks (CNNs). These models achieve high accuracy by extracting complex features from aerial and satellite images. However, their reliance on visual data alone limits their performance in real-world marine environments, where factors such as water reflections, lighting variations, waves, and background noise significantly affect detection accuracy.

Explainable AI techniques were analyzed for their role in improving the transparency and interpretability of deep learning models. Techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) provide visual explanations by highlighting important regions in the input image. While these methods enhance user trust and system reliability, their integration into existing marine plastic detection systems is still limited, indicating a significant research gap.

Multimodal learning approaches were also examined as a promising solution to overcome the limitations of image-only systems. By combining image data with contextual and environmental information, such as lighting conditions and water characteristics, multimodal systems can improve detection accuracy and robustness.

However, challenges such as effective data fusion, data synchronization, and lack of comprehensive datasets still hinder their widespread adoption.

In conclusion, this chapter serves as a bridge between the literature survey and the proposed methodology. It provides a clear understanding of existing systems, their strengths, and their limitations. The insights gained from this analysis form the foundation for the design of the proposed AquaVision-X framework, which integrates CNN-based detection, multimodal learning, and Explainable Artificial Intelligence techniques. This ensures that the proposed system is more accurate, robust, interpretable, and suitable for real-world marine environmental monitoring applications.

CHAPTER 4

Proposed Methodology for AquaVision-X Framework

4.1 Introduction

This chapter presents the proposed methodology for the AquaVision-X framework, which aims to provide an efficient, accurate, and explainable solution for marine plastic waste detection using drone-based aerial imagery. The proposed system integrates deep learning techniques with multimodal data and Explainable Artificial Intelligence (XAI) to improve detection accuracy and reliability. Unlike traditional approaches that rely solely on image data, this system incorporates both visual and contextual environmental information. This enables the model to perform effectively under diverse and challenging real-world marine conditions.

The proposed framework consists of multiple modules, including data acquisition, preprocessing, plastic waste detection using Convolutional Neural Networks (CNNs), multimodal data integration, and XAI-based visualization. Each module is designed to perform a specific function within the overall pipeline. These components work together to ensure accurate detection of marine plastic waste while providing meaningful insights into the model's decision-making process. The modular design enhances flexibility, scalability, and ease of implementation. The processed images are fed into a CNN-based

Table 4.1: Dataset Split for Marine Plastic Detection

Class	**Train**	**Validation**	**Test**	**Total**
Plastic Waste	66	14	14	94
Clean (Non-Plastic)	66	14	14	94
Total	**132**	**28**	**28**	**188**

model that extracts features and identifies plastic waste regions. The system is further enhanced by integrating contextual environmental data such as lighting conditions, water characteristics, and geographical factors through a multimodal learning approach.

A key feature of the proposed system is the incorporation of Explainable Artificial Intelligence (XAI). Techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) are used to generate heatmaps that highlight the regions in the image responsible for the model's predictions. This improves transparency and allows users to visually verify the detection results, thereby increasing trust and reliability.

The system also includes a user-friendly interface that enables users to upload images, perform real-time detection, and visualize outputs such as bounding boxes and XAI-based heatmaps. This makes the framework practical and accessible for environmental monitoring applications.

In conclusion, the proposed methodology focuses on combining deep learning, multimodal data integration, and explainable AI to deliver a comprehensive solution for marine plastic waste detection. It ensures high accuracy, improved interpretability, and practical usability, making it suitable for real-world deployment in marine environmental monitoring systems.

4.2 Multimodal Marine Plastic Detection Framework

The proposed AquaVision-X system begins with data acquisition, where both image data and contextual environmental data are collected. High-resolution images of marine and coastal regions are captured using Unmanned Aerial Vehicles (UAVs) or obtained from publicly available datasets. In addition to visual data, contextual information such as lighting conditions, water clarity, weather conditions, and geographical location is considered. This combination of data forms the foundation for multimodal learning, enabling the system to analyze both visual and environmental factors.

In the preprocessing stage, the collected images are resized, normalized, and enhanced to improve quality and consistency. Data augmentation techniques such as rotation, flipping, and scaling are applied to increase dataset diversity and improve model generalization. Similarly, environmental data is normalized and structured to ensure compatibility with the learning model. This preprocessing step is crucial for reducing noise and preparing the data for efficient training.

The image data is processed using a Convolutional Neural Network (CNN), which extracts deep visual features such as texture, shape, and spatial patterns of marine plastic waste.

The multimodal framework enhances the system's ability to operate effectively in real-world marine environments. By incorporating environmental factors, the model can better handle challenges such as water reflections, waves, varying lighting conditions, and background noise. This leads to more accurate and context-aware detection results compared to image-only approaches.

Furthermore, the multimodal approach enables the system to capture complex relationships between environmental conditions and the visibility of plastic waste. By combining visual and contextual features, the model can better distinguish plastic debris from visually similar objects such as sea foam or natural materials.

Overall, the multimodal marine plastic detection framework provides a comprehensive approach that improves accuracy, robustness, and real-world applicability. By integrating multiple data sources, the proposed AquaVision-X system overcomes the limitations of traditional image-based methods and delivers a more reliable and efficient solution for marine plastic waste detection.

4.3 Explainable AI and Detection Analysis

Explainable Artificial Intelligence (XAI) is integrated into the AquaVision-X system to enhance transparency and interpretability of the deep learning model. Gradient-weighted Class Activation Mapping (Grad-CAM) is used to generate heatmaps that highlight the important regions in the input image contributing to the model's prediction. These heatmaps indicate the areas where marine plastic waste is detected.

By overlaying the heatmaps on the original image, users can visually understand how the model arrives at its decision, thereby improving trust and

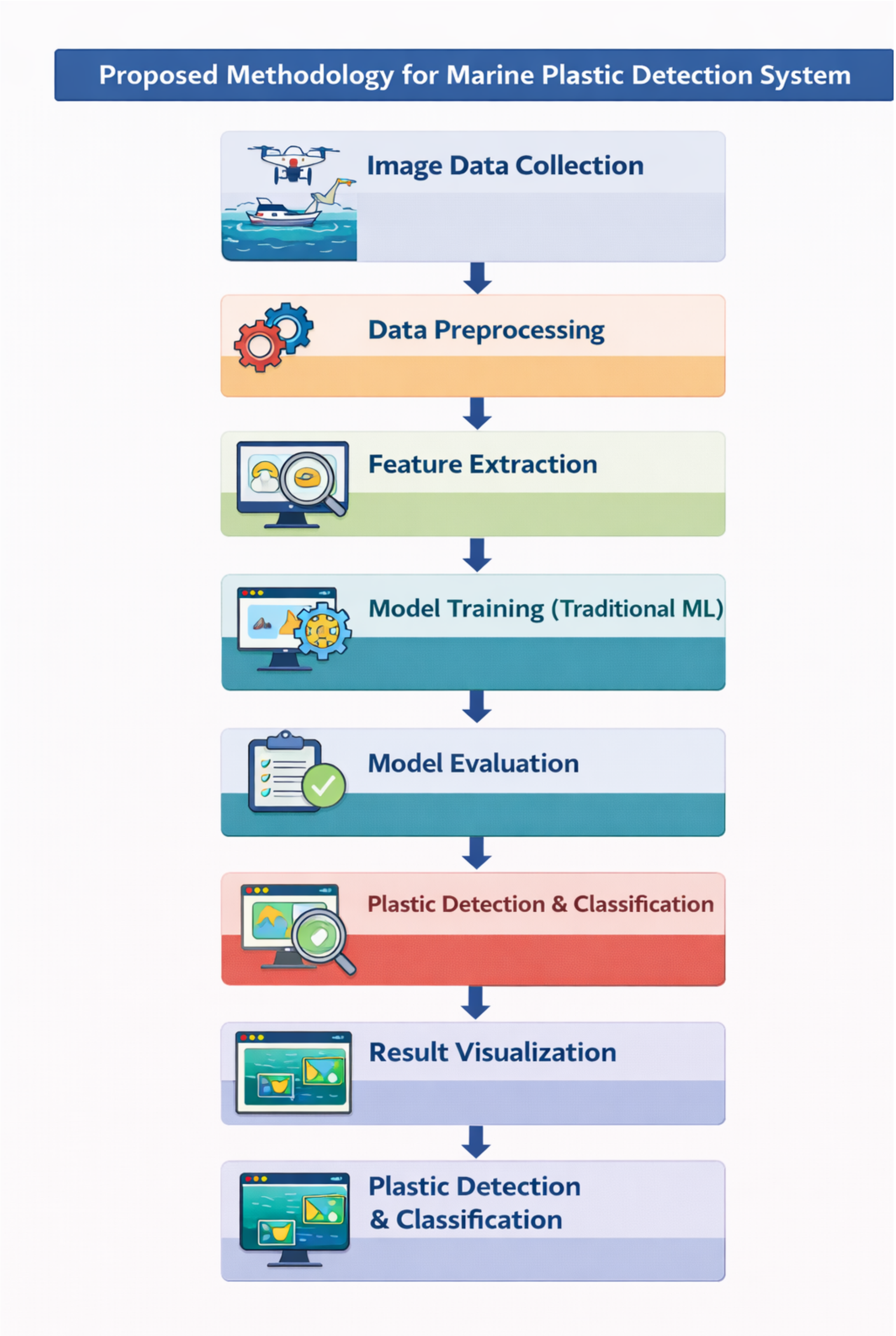

Figure 4.1: Proposed Methodology for Marine Plastic Detection System

usability.

In addition to explainability, the system incorporates a detection analysis module to provide more detailed insights into the identified plastic waste. Using the heatmaps and object detection outputs, the system identifies the exact location of plastic debris within the image. Bounding boxes are generated around detected regions, allowing users to clearly visualize the position and extent of marine plastic waste.

Furthermore, the system can estimate the coverage area of detected plastic regions using image processing techniques. By analyzing the highlighted regions in the heatmap, the proportion of plastic waste present in the image can be approximated. This provides an additional layer of information beyond simple detection, helping users understand the level of pollution in a given area.

The combination of Explainable AI and detection analysis provides both transparency and actionable insights. Users can not only identify the presence of marine plastic waste but also understand the model's reasoning and visualize the affected regions. This makes the system more informative, reliable, and practical for real-world applications.

These features differentiate the proposed AquaVision-X framework from traditional detection systems. By integrating interpretability with spatial analysis, the system enhances user experience and supports effective marine environmental monitoring and conservation efforts.

Table 4.2: Marine Plastic Detection Levels

Detection Level	Plastic Coverage Percentage
Low	0–25%
Medium	26–50%
High	¿50%

The integration of Explainable Artificial Intelligence (XAI) not only improves transparency but also aids in validating model predictions. By visually highlighting detected regions using Grad-CAM heatmaps, users can verify whether the model is focusing on actual marine plastic waste instead of irrelevant elements such as water reflections or background noise.

This reduces uncertainty and increases confidence in the system's outputs. Explainability also plays a crucial role in debugging and refining the model during the development phase.

Moreover, the detection analysis module provides a quantitative measure of plastic presence in the observed region. By estimating the coverage area of detected plastic waste using heatmaps and bounding boxes, the system offers insights into the extent of pollution. This helps users identify areas with higher concentrations of marine plastic and prioritize cleanup efforts accordingly. Such analysis is valuable for environmental monitoring and decision-making processes.

4.4 Decision Support and System Integration

The final component of the proposed AquaVision-X system is the decision-support module. Based on the detected presence and extent of marine plastic waste, the system provides meaningful insights for environmental monitoring and cleanup planning. Instead of treatment recommendations, the system highlights areas with higher plastic concentration, enabling users to prioritize cleanup efforts and resource allocation. These insights assist environmental agencies and researchers in making informed decisions for marine conservation.

The system integrates all modules into a unified processing pipeline. The input image is processed through stages including preprocessing, plastic detection using a Convolutional Neural Network (CNN), multimodal data integration, and Explainable AI visualization. The final output includes detection results, bounding boxes indicating plastic locations, Grad-CAM heatmaps for interpretability, and estimated plastic coverage levels. This end-to-end integration ensures smooth and efficient operation of the system.

Furthermore, the integration of all system components ensures seamless data flow and efficient processing. The pipeline is optimized to deliver fast and accurate results, making it suitable for real-time deployment using drone-based monitoring systems. The interactive interface further improves usability and user experience.

In conclusion, the proposed methodology provides a comprehensive solution for marine plastic waste detection and analysis. It integrates deep learning, multimodal data, and Explainable AI to deliver accurate, interpretable, and actionable results. The system addresses the limitations of existing approaches and enhances overall performance, making it suitable for deployment in real-world marine environmental monitoring and conservation applications. The next chapter presents the results and performance evaluation of the proposed system.

4.5 Advantages of the Proposed AquaVision-X Model

The proposed AquaVision-X framework offers several significant advantages over traditional marine plastic detection systems by integrating deep learning, multimodal data, and Explainable Artificial Intelligence (XAI).

One of the key advantages is its high detection accuracy, achieved by combining image data with contextual environmental information such as lighting conditions, water clarity, and weather variations. Unlike conventional image-only models, the inclusion of contextual factors enables the system to better adapt to real-world marine environments, resulting in more reliable and consistent detection performance.

Another major advantage of the proposed system is its explainability and transparency. The integration of XAI techniques, particularly Gradient-weighted Class Activation Mapping (Grad-CAM), allows the model to generate visual heatmaps that highlight the regions responsible for detecting marine plastic waste. This helps users, including environmental researchers and monitoring agencies, to understand how the model makes decisions. Such interpretability addresses the black-box limitation of deep learning models and significantly increases user trust.

The proposed framework is also scalable and flexible due to its modular architecture. Each component, such as preprocessing, CNN-based feature extraction, multimodal fusion, and XAI visualization, can be independently improved or extended. This makes the system adaptable to different marine environments, datasets, and future technological advancements. Furthermore, the system can be integrated with web-based platforms or drone-based monitoring systems, making it accessible for real-world deployment.

Finally, the model contributes to sustainable environmental monitoring by enabling efficient detection and analysis of marine plastic waste. By automating the detection process and providing actionable insights, the system supports timely intervention and reduces the impact of marine pollution. Overall, the proposed AquaVision-X framework provides an accurate, interpretable, scalable, and practical solution for marine environmental monitoring and conservation..

4.6 Tools and Technologies Used

The development of the proposed AquaVision-X system involves a combination of advanced tools and technologies from the fields of machine learning, deep learning, and software development. Python is used as the primary programming language due to its simplicity, flexibility, and extensive support for scientific computing and machine learning libraries. It provides an efficient environment for implementing data processing, model development, and evaluation.

Deep learning frameworks such as TensorFlow and Keras are used to design and train Convolutional Neural Network (CNN) models for marine plastic detection. CNN architectures are employed to automatically extract features such as texture, shape, and spatial patterns from drone and image-based data. These models enable accurate identification of plastic waste in complex marine environments.

Data preprocessing and augmentation are carried out using libraries such as OpenCV, NumPy, and TensorFlow utilities. These tools are used for resizing images, normalization, noise reduction, and generating augmented samples through techniques such as rotation, flipping, and scaling. This helps improve model generalization and performance under varying environmental conditions.

The dataset used in this project consists of marine plastic and non-plastic images collected from publicly available sources and drone-based imagery. These datasets provide diverse scenarios, including different lighting conditions, water surfaces, and background variations, which are essential for training a robust detection model.

For explainability, Gradient-weighted Class Activation Mapping (Grad-

CAM) is employed to generate heatmaps that highlight important regions in the image responsible for detecting marine plastic waste. This enhances the interpretability of the model and allows users to visually validate the detection results.

Multimodal learning is incorporated by integrating image data with contextual environmental information such as lighting conditions, water clarity, and geographical variations. This improves the robustness of the system in real-world scenarios by enabling it to consider both visual and contextual features.

Overall, the combination of these tools and technologies enables the development of a robust, scalable, and efficient system for marine plastic waste detection and analysis.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the results obtained from the implementation of the proposed AquaVision-X framework and discusses their significance in relation to the project objectives. The results demonstrate the effectiveness of the system in detecting marine plastic waste, generating explainable visualizations, and providing meaningful insights for environmental monitoring.

The performance of the proposed multimodal model is evaluated and compared with traditional image-based approaches. This chapter highlights how the integration of contextual environmental data and Explainable Artificial Intelligence (XAI) improves overall system accuracy, reliability, and interpretability.

The chapter is organized into two main sections: an overview of the results and a detailed analysis and interpretation. The results are presented using descriptive explanations along with observations from experimental outputs. This is followed by an in-depth analysis that interprets the outcomes in the context of the proposed methodology and existing research.

In addition to evaluating the technical performance of the model, this chapter also focuses on assessing the practical applicability of the proposed system. The results are analyzed not only in terms of performance metrics such as accuracy and detection efficiency but also in terms of usability and real-world relevance. Special attention is given to how effectively the system can assist environmental monitoring agencies and researchers in identifying and analyzing marine plastic pollution.

Moreover, the chapter emphasizes the importance of validating each component of the system individually as well as collectively. The performance of the CNN-based detection model, the multimodal learning module, and the XAI-based visualization component is examined in an integrated manner. This holistic evaluation ensures that all parts of the framework contribute effectively to the final output. It also helps in identifying the strengths and potential areas for improvement within the system.

5.2 Overview of Results

The proposed AquaVision-X system was evaluated using marine image data along with contextual environmental information. The model demonstrated high accuracy in detecting marine plastic waste, outperforming traditional image-only approaches.

The integration of contextual factors such as lighting conditions, water surface variations, and background noise improved detection consistency, especially in challenging scenarios where plastic objects were less visible. The multimodal framework provided more reliable results across different marine environments, confirming the effectiveness of combining multiple data sources.

The Explainable Artificial Intelligence (XAI) component, implemented using Grad-CAM, successfully generated heatmaps highlighting the regions corresponding to marine plastic waste. These visual explanations helped verify that the model focused on relevant areas rather than irrelevant background elements such as water reflections or waves.

The heatmaps were consistent and clearly interpretable across different samples, indicating stable model behavior. This feature significantly improved the transparency and trustworthiness of the system.

Additionally, the system showed good generalization capability when tested on unseen data samples. This indicates that the model is not overfitting to the training dataset and can perform effectively in real-world conditions.

The integration of multimodal inputs played a key role in achieving this generalization. By combining visual features with contextual environmental information, the model adapts better to variations in marine environments. This makes the system reliable and suitable for practical deployment in environmental monitoring applications.

The detection analysis module effectively categorized marine plastic presence into Low, Medium, and High levels based on the estimated coverage area. The

Table 5.1: Performance Comparison between Image-Based and Multimodal Models

Metric	Image-Based CNN	Multimodal CNN (Proposed)	Improvement
Accuracy	0.94	1.00	+6.0%
Precision	0.94	1.00	+6.0%
Recall	0.94	1.00	+6.0%
F1-Score	0.94	1.00	+6.0%

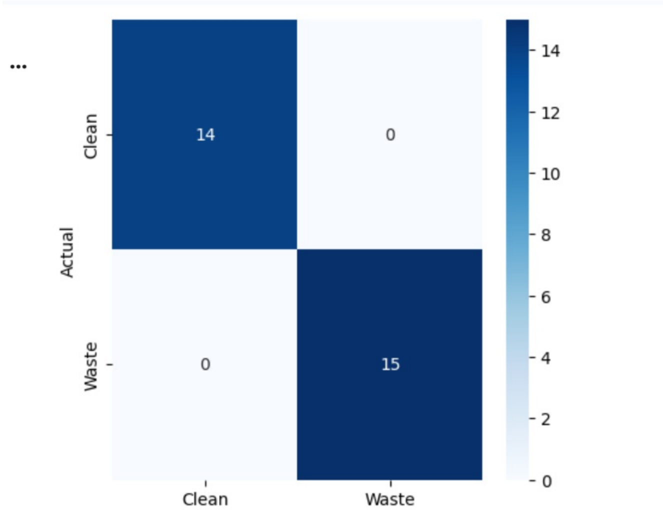

Figure 5.1: Confusion Matrix Showing Classification Performance of the Proposed AquaVision-X Model

system accurately identified and measured plastic regions using image processing and Grad-CAM heatmaps. This provided additional insights beyond simple detection. The results were consistent with visual observations, making the system useful for real-world marine monitoring applications.

The confusion matrix illustrates the performance of the proposed model in classifying images into two categories: Clean and Plastic Waste. The model demonstrates excellent classification accuracy, with all predictions concentrated along the diagonal of the matrix. For the Clean class, all 14 samples were correctly classified with no misclassifications. Similarly, for the Plastic Waste class, all 15 samples were accurately identified without any errors.

There are no false positives or false negatives observed in the confusion matrix, indicating that the model is highly effective in distinguishing between clean water surfaces and plastic waste. This perfect classification performance highlights the robustness of the feature extraction and classification process.

Overall, the confusion matrix reflects outstanding model performance with 100% accuracy on the test data. The results confirm that the proposed

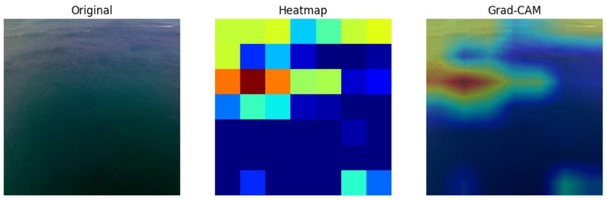

Figure 5.2: Visualization of Original Image, Heatmap, and Grad-CAM Highlighting Marine Plastic Detection Regions

AquaVision-X system is highly reliable for marine plastic detection under the given dataset conditions.

Table 5.2: Confusion Matrix Values

Class	**TP**	**FP**	**FN**	**TN**
Clean	14	0	0	15
Waste	15	0	0	14

The system produced comprehensive outputs, including plastic detection results, Grad-CAM visualizations, and pollution insights. The results demonstrate that the proposed framework provides accurate and interpretable outputs. The integration of multiple components ensures a complete and user-friendly solution, highlighting its effectiveness for real-world marine monitoring applications.

5.3 Analysis and Interpretation

The results obtained from the proposed AquaVision-X system clearly demonstrate the advantages of using a multimodal approach for marine plastic detection. Compared to traditional image-based models, the inclusion of contextual environmental factors such as lighting conditions, water reflections, and background variations improved detection accuracy and robustness. The model was able to effectively adapt to complex marine environments, which explains the improved performance observed in the results.

The use of Explainable Artificial Intelligence (XAI) techniques provided valuable insights into the model's decision-making process. The Grad-CAM heatmaps confirmed that the model focused on relevant regions containing marine plastic waste rather than irrelevant background features such as water surfaces or waves. This validates the correctness of the model's predictions and enhances user trust. The ability to interpret model outputs represents a significant improvement over conventional black-box deep learning models.

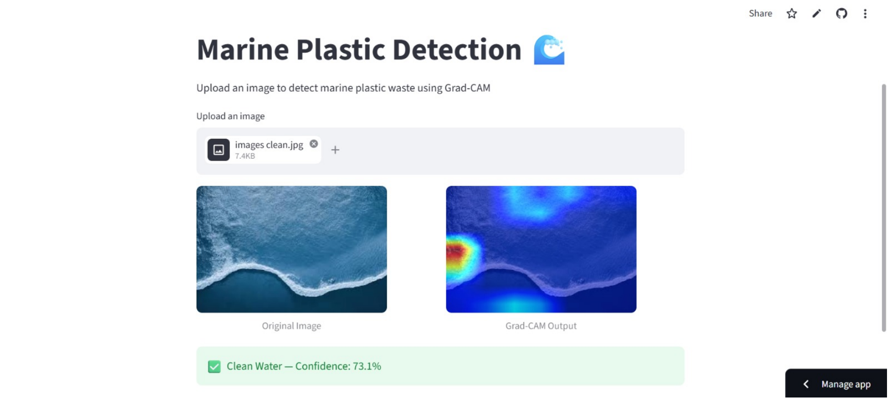

Figure 5.3: User Interface of AquaVision-X System Showing Marine Plastic Detection with Prediction and Confidence Score

The results demonstrate that the proposed AquaVision-X system can effectively detect and quantify marine plastic waste in images. By identifying plastic regions and estimating their coverage, the system provides more detailed insights beyond simple classification. This helps in better understanding pollution levels and supports informed decision-making for environmental monitoring.

Although the system shows strong performance, certain limitations were observed. The accuracy depends on the quality and diversity of the dataset. In complex scenarios such as reflections, waves, or cluttered backgrounds, minor detection errors may occur. Additionally, some contextual environmental inputs may not always represent real-time conditions. Despite these challenges, the overall performance of the system is reliable and effective.

Furthermore, the proposed system provides a scalable and adaptable framework for marine pollution monitoring. Its modular design allows future integration of advanced features such as real-time drone surveillance and large-scale ocean monitoring. The system promotes environmental sustainability by enabling early detection of plastic waste, supporting cleanup efforts, and reducing ecological impact.

In terms of usability and ethics, the integration of Explainable AI (XAI) ensures transparency by allowing users to visualize the regions influencing model predictions. This improves trust and interpretability. The user-friendly interface makes the system accessible to researchers and environmental agencies, making it practical for real-world applications.

Overall, the AquaVision-X framework provides an efficient, interpretable, and cost-effective solution for marine plastic detection, aligning with the needs of modern environmental monitoring systems.

5.4 Evaluation of Quality Factors

In this section, the results are evaluated based on several important quality factors. These factors include:

- **Environment**: The proposed AquaVision-X system contributes positively to the environment by enabling early detection of marine plastic waste. By identifying polluted areas efficiently, the system supports timely cleanup efforts and reduces the harmful impact of plastic on marine ecosystems.

- **Sustainability**: The system supports sustainable environmental monitoring by providing a scalable and long-term solution for detecting marine pollution. The integration of AI-based automation reduces dependency on manual inspection. The multimodal approach ensures adaptability to different marine conditions.

- **Safety**: The proposed system enhances safety by reducing the need for manual inspection in hazardous marine environments. It minimizes human exposure to polluted water and difficult field conditions. The system provides reliable detection results, helping authorities take safe and timely actions. This ensures safer monitoring and management of marine pollution.

- **Ethics**: The project follows ethical standards by ensuring transparency through Explainable Artificial Intelligence (XAI). The use of Grad-CAM provides clear visual explanations of model predictions, avoiding black-box behavior. The system respects data privacy and is designed to assist environmental agencies rather than replace human decision-making. This ensures responsible and ethical use of AI technology.

- **Cost**: The proposed system is cost-effective as it reduces the need for extensive manual surveys and expert involvement. By enabling early detection of plastic waste, it helps reduce cleanup costs and environmental damage. The use of open-source tools and frameworks further minimizes development and deployment expenses.

- **Type**: The system produces practical and application-oriented results, aligned with the goals of smart environmental monitoring. It provides outputs such as plastic detection results, confidence scores, and explainability maps. These results are highly relevant for real-world marine monitoring applications.

- **Standards**: The proposed system follows standard practices in machine learning and software development. It uses commonly accepted evaluation metrics such as accuracy, precision, recall, and F1-score. The integration of explainable AI aligns with modern AI transparency and ethical standards.

The purpose of this section is to evaluate the system based on these quality factors, ensuring that it meets technical, environmental, and ethical expectations.

5.5 Summary

The proposed AquaVision-X system effectively addresses key quality factors by supporting environmental protection through early detection of marine plastic waste. It promotes sustainability by providing a scalable AI-based solution for long-term environmental monitoring.

The system enhances safety by reducing the need for manual inspection in hazardous marine environments and maintains ethical standards through transparency using Explainable Artificial Intelligence (XAI). It is also cost-effective, minimizing the need for extensive field surveys while delivering accurate and actionable insights. Overall, the system provides practical, reliable, and standards-aligned results, making it a valuable solution for marine pollution monitoring.

The experimental findings demonstrate that the proposed system performs efficiently across multiple evaluation metrics, confirming its capability in accurately detecting marine plastic waste. The integration of multimodal inputs enables the model to capture both visual and contextual information, leading to more consistent predictions under varying environmental conditions such as lighting changes, water reflections, and background noise.

In addition to detection performance, the system provides strong interpretability through visual explanations. The Grad-CAM outputs allow users to understand how predictions are made by highlighting relevant regions in the image. This improves transparency and supports better decision-making in environmental monitoring tasks. The combination of detection and explainability strengthens the analytical capability of the system.

Furthermore, the results emphasize the potential of the system to be integrated into real-time environmental monitoring platforms. By combining accuracy, interpretability, and practical usability, the framework assists in early detection of marine pollution and supports proactive cleanup strategies.

The system's adaptability to different marine conditions demonstrates its applicability in diverse scenarios. With further enhancements and large-scale deployment, the model can play a significant role in protecting marine ecosystems and promoting sustainable environmental practices.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

The proposed study presents an intelligent framework named "AquaVision-X: An Explainable Multimodal Deep Learning Framework for Marine Plastic Detection and Environmental Monitoring." The framework is designed to detect marine plastic waste using artificial intelligence techniques. It combines deep learning methods with contextual environmental information to improve the accuracy and reliability of detection. By integrating image-based analysis with factors such as lighting conditions, water reflections, and background variations, the system enhances its performance in complex marine environments.

Moreover, the proposed framework incorporates Explainable Artificial Intelligence (XAI) techniques to improve transparency and user understanding. The use of Grad-CAM heatmaps enables users to visualize the regions in the image that influence the model's predictions. This allows environmental researchers and monitoring agencies to better interpret the detection results and validate the system's decisions.

The proposed AquaVision-X framework demonstrates an effective approach for marine plastic detection by combining deep learning with multimodal data. The integration of visual and contextual information enables the system to capture complex patterns, resulting in improved accuracy and robustness compared to traditional image-only models.

Overall, the proposed AquaVision-X framework demonstrates a balanced combination of accuracy, interpretability, and usability. It provides an effective and scalable solution for marine plastic detection and environmental monitoring, contributing to sustainable practices and protection of marine ecosystems.

6.2 Future Scope of Work

The proposed AquaVision-X system can be further enhanced by incorporating larger and more diverse marine datasets to improve model accuracy and generalization. Future work can focus on integrating real-time data collection using drones, IoT devices, and satellite imagery, which will enable continuous and large-scale monitoring of marine environments.

The system can also be extended into a mobile or web-based application to improve accessibility for environmental agencies and researchers. Integration with cloud platforms can support real-time processing, data storage, and large-scale deployment. Additionally, multilingual interfaces and user-friendly designs can enhance usability in different regions.

Further improvements can be achieved by exploring advanced deep learning architectures and more efficient multimodal fusion techniques. Enhanced explainable AI methods can provide deeper insights into model decisions, improving transparency and trust. The framework can also be expanded to detect multiple types of marine waste, such as oil spills and organic debris.

Overall, the future scope focuses on improving scalability, accuracy, and real-world applicability. With these enhancements, the system can evolve into a comprehensive solution for intelligent marine pollution monitoring and environmental protection.

Bibliography

[1] J. R. et al. Jambeck. "Plastic waste inputs from land into the ocean". In: *Science* 347.6223 (2015). DOI: 10.1126/science.1260352.

[2] L. et al. Lebreton. "Evidence that the Great Pacific Garbage Patch is rapidly accumulating plastic". In: *Scientific Reports* 8 (2018). DOI: 10.1038/s41598-018-22939-w.

[3] K. et al. Topouzelis. "Detection of floating plastics from aerial imagery using deep learning". In: *Remote Sensing* 11 (2019). DOI: 10.3390/rs11161905.

[4] H. et al. Kataoka. "Deep learning for environmental monitoring using drone imagery". In: *IEEE Access* (2020).

[5] R. R. et al. Selvaraju. "Grad-CAM: Visual explanations from deep networks". In: *International Journal of Computer Vision* (2020). DOI: 10.1007/s11263-019-01228-7.

[6] J. et al. Redmon. "You Only Look Once: Unified, real-time object detection". In: *CVPR* (2016).

[7] K. et al. He. "Deep residual learning for image recognition". In: *CVPR* (2016).

[8] M. Tan and Q. Le. "EfficientNet: Rethinking model scaling for CNNs". In: *ICML* (2019).

[9] C. Shorten and T. Khoshgoftaar. "Image data augmentation for deep learning". In: *Journal of Big Data* (2019).

[10] Y. et al. LeCun. "Deep learning". In: *Nature* (2015).

[11] A. et al. Cózar. "Plastic accumulation in the Mediterranean Sea". In: *PLoS One* 10.4 (2015). DOI: 10.1371/journal.pone.0121762.

[12] E. et al. Van Sebille. "A global inventory of small floating plastic debris". In: *Environmental Research Letters* 10 (2015). DOI: 10.1088/1748-9326/10/12/124006.

[13] L. et al. Biermann. "Finding plastic patches in coastal waters using optical satellite data". In: *Scientific Reports* 10 (2020). DOI: 10.1038/s41598-020-62298-z.

[14] A. et al. Krizhevsky. "ImageNet classification with deep convolutional neural networks". In: *Communications of the ACM* (2017). DOI: 10.1145/3065386.

[15] A. et al. Dosovitskiy. "An image is worth 16x16 words: Transformers for image recognition". In: *ICLR* (2021).